

Landmark Engines

By the outbreak of World War I, two basic types of aero engine were in production. Water-cooled engines generally had individual cylinders mounted in line on an aluminum alloy crankcase. There were a few air-cooled inline engines, but limited knowledge of effective cylinder head design favored the seemingly mad idea of spinning the entire engine to keep it cool. Thirsty and requiring frequent servicing, these “rotary” engines were, nevertheless, light and powerful. After World War I, simpler air-cooled “sixes,” “fours,” and “twins” of various configurations were used in a new class of light aircraft that emerged in the 1920s.

While its complexity and weight counted against it in the civil market, the slender shape of the liquid-cooled inline engine

was reckoned to offer less air resistance and greater speed, and for these reasons it was mostly used in racers and military aircraft.

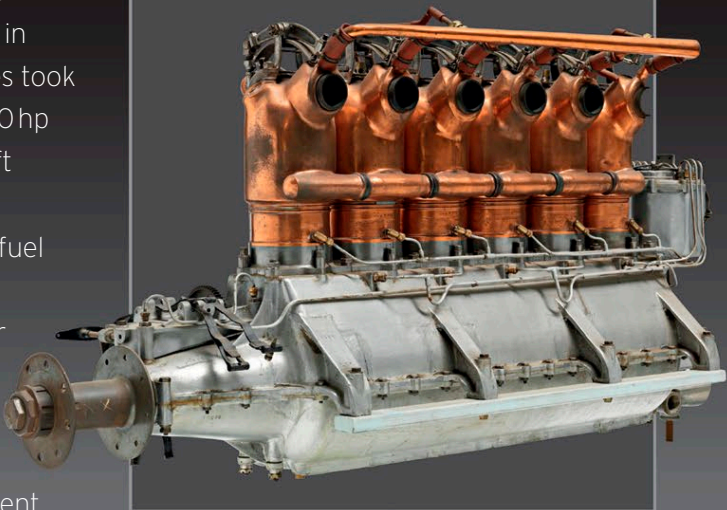
Large piston engines were superseded in the late 1940s by turbojet engines, and eventually the turboprops and modern turboprops that were used in airliners. Gradually, turbine engines took over all applications above the 300 hp mark, leaving only the light aircraft market to piston engines.

In recent years the high cost of fuel has made heavier but more economical diesel engines popular for training and touring aircraft.

Today, even “greener” alternative power units using electric motors and fuel cells are under development.

AUSTRO-DAIMLER 6	
Dates produced	1910-1916
Displacement	851 cu in (13.9 liters)
Maxi power output	154 hp at 1,400 rpm

One of the first successful inline, water-cooled aero engines developed in Europe, the Austro-Daimler 6 was designed by Dr. Ferdinand Porsche. The most efficient and reliable aero engine at the outbreak of WWI, its overhead-cam layout was widely copied. The engine was also built under licence by other companies, including William Beardmore & Co Ltd in Scotland. This British-built example was fitted to the FE series pusher biplanes operated by the RFC.



CONTINENTAL A-40	
Dates produced	1931-1938
Displacement	115 cu in (1.9 liters)
Max power output	40 hp at 2,550 rpm

The Continental A-40 was a deeply flawed engine that nevertheless established the pattern for most light aircraft engines since. The first A-40s overheated because the cylinder head was insulated from the cylinders by a gasket that blocked heat flow, and the side exhaust valves were inadequately cooled. The crankshaft tended to break because the thrust bearing was placed in error at the rear of the block. These flaws were eliminated in the A-50 and A-65 that followed.



ROLLS-ROYCE MERLIN III	
Dates produced	1936-1950 (all Marks)
Displacement	1,647 cu in (27.0 liters)
Max power output	1,440 hp at 3,000 rpm/5,500 ft

The Rolls-Royce Merlin has been called unkindly “a triumph of development over design.” Certainly its basic architecture was not as robust as the US Allison V12, nor was its detail design as impressive as its German rival, the DB601. However, thanks to the perseverance of Rolls-Royce’s engineers in general, and the brilliance of the supercharger development team in particular, the Merlin’s power output was doubled while maintaining its outstanding reliability.



BRISTOL CENTAURUS	
Dates produced	1943 onward
Displacement	3,270 cu in (53.6 liters)
Max power output	2,550 hp at 4,000 ft (MkXVIII)

Having used conventional “penny on a stick” poppet valves in his early designs, chief engineer Roy Fedden turned to sleeve valves for Bristol’s later and more powerful radial engines. The Centaurus was the ultimate development, although its gestation was slow: work on what was envisaged as an engine for “future aircraft projects” began in 1937, the demanding 2,000 hp type test was passed in 1939, and it did not go into production until 1943. Postwar models produced over 3,000 hp.

